Business Questions and Visualization Report

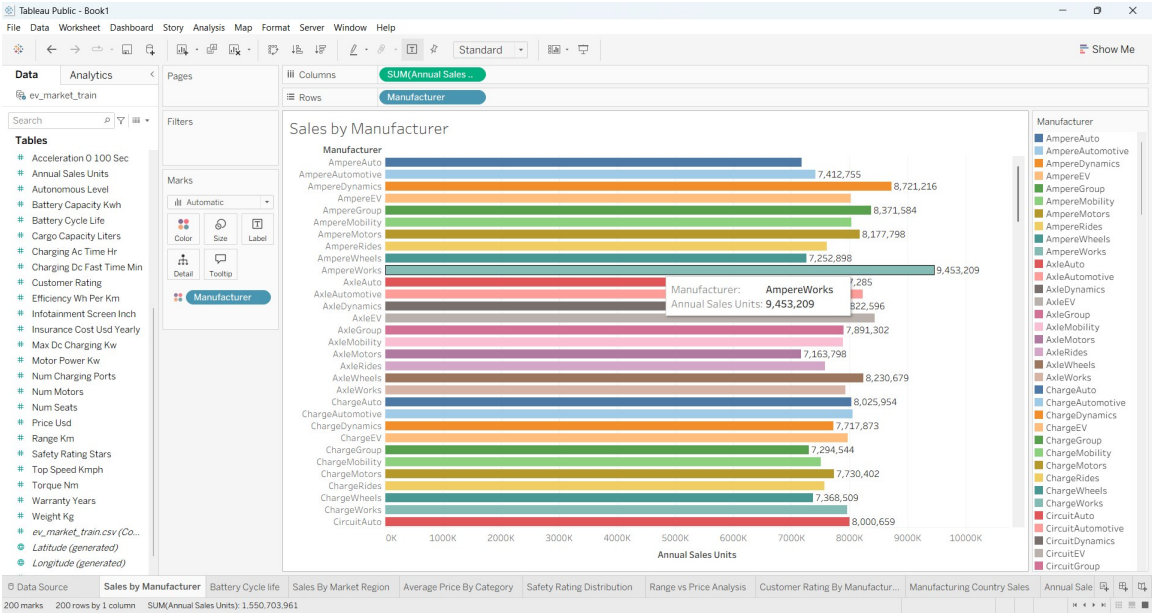
Field	Details
Date	02 November 2026
Team ID	N/A (Individual Project)
Project Name	EV Value & Longevity Analytics Dashboard
Prepared by	Piyush Malviya
Maximum Marks	5 Marks

Good visualisations start with good business questions. Each question below is tied to one Tableau worksheet, the chart type used to answer it, and the finding from the training data (200,000 vehicles, 200 manufacturers).

1. Which manufacturers lead in total annual sales?

Visualization: Horizontal bar chart of SUM(Annual Sales Units) by Manufacturer, coloured by manufacturer.

Finding: AmpereWorks leads with 9.45M units, then CircuitWorks (8.92M) and MomentumMobility (8.78M). The smallest of the 200 is DynamoAutomotive (6.32M), so no single OEM dominates.

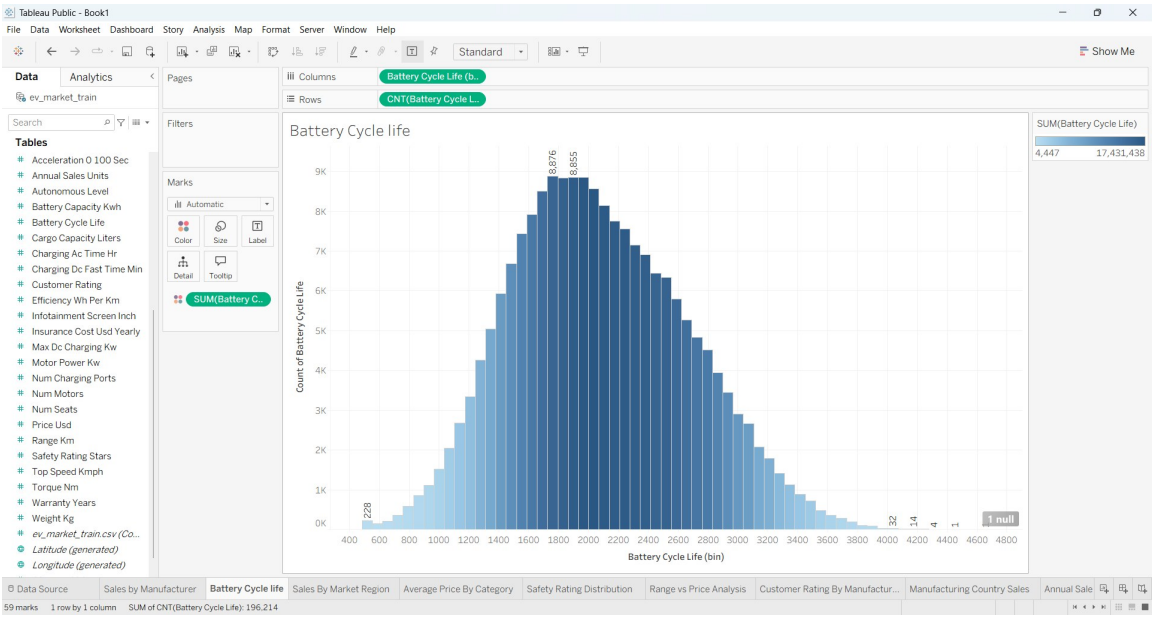


Sales by Manufacturer (Tableau worksheet)

2. How long do EV batteries last, and how are cycle-life values distributed?

Visualization: Histogram of Battery Cycle Life using 100-cycle bins, with colour showing density.

Finding: The median is 2,031 cycles, the middle 90% of vehicles fall between 1,173 and 3,101 cycles, and the tallest bins sit around 1,800 to 2,000 cycles.

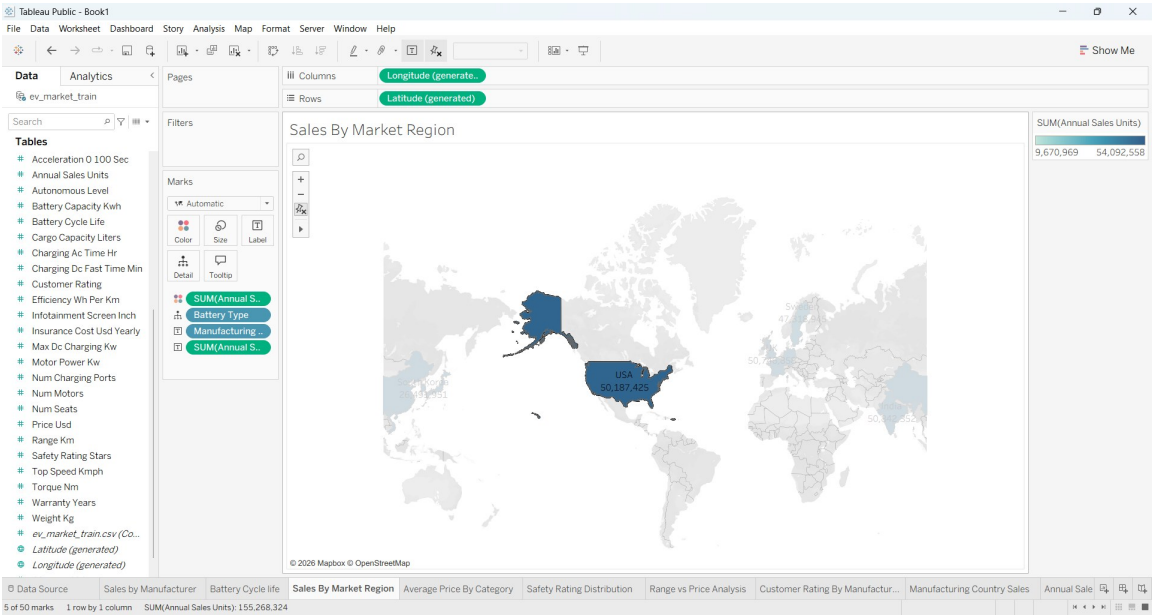


Battery Cycle Life distribution (Tableau worksheet)

3. Which regions and countries carry the most sales volume?

Visualization: Filled map of sales, with country and battery type in the tooltip.

Finding: Seven market regions are almost level, from Europe at 217.5M to South Asia at 224.0M units, a spread of about 3%.

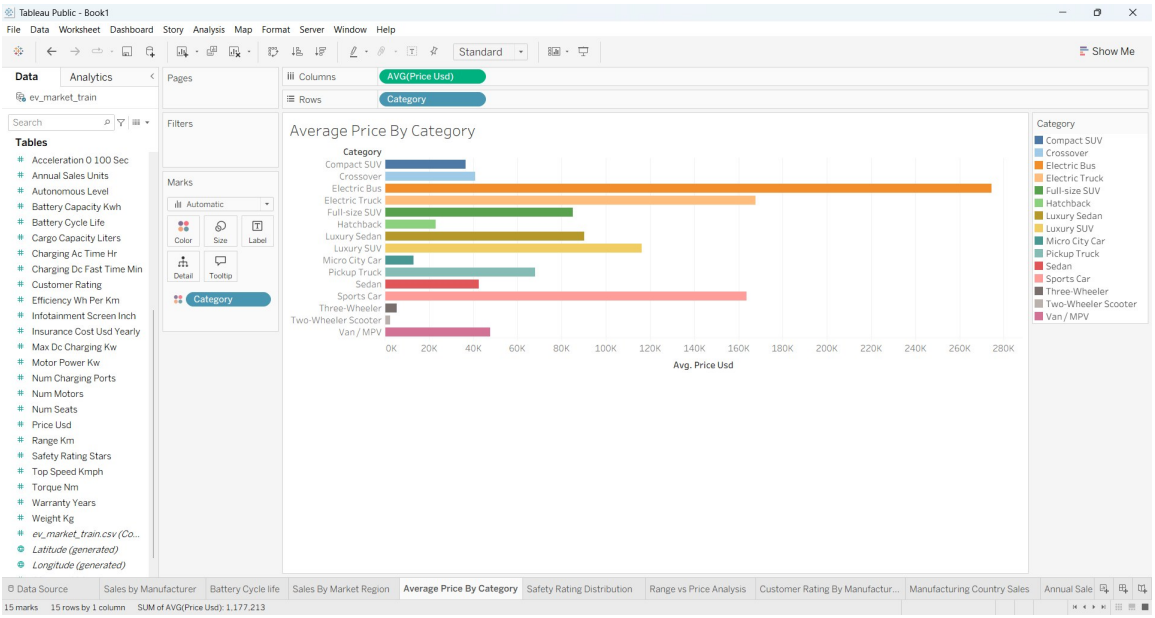


Sales by Market Region (Tableau worksheet)

4. How does average price change across the 15 vehicle categories?

Visualization: Horizontal bar chart of AVG(Price USD) by Category.

Finding: Electric Bus is the most expensive at about $274.5K, then Electric Truck ($167.7K) and Sports Car ($163.6K). Two-Wheeler Scooters average only $2.6K.

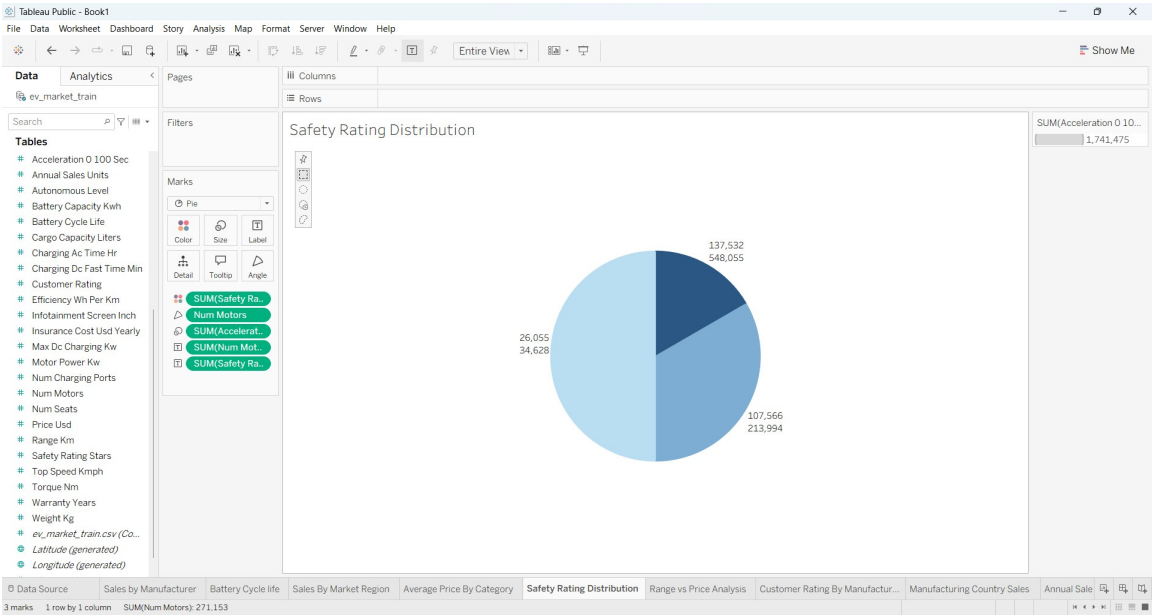


Average Price by Category (Tableau worksheet)

5. How are safety ratings distributed across the market?

Visualization: Pie chart of vehicle count by Safety Rating (Stars).

Finding: 4-star vehicles are 52.5% of records, 5-star 23.8% and 3-star 22.2%. Only 1.6% are rated 2 stars or lower. The average is 3.98 stars.

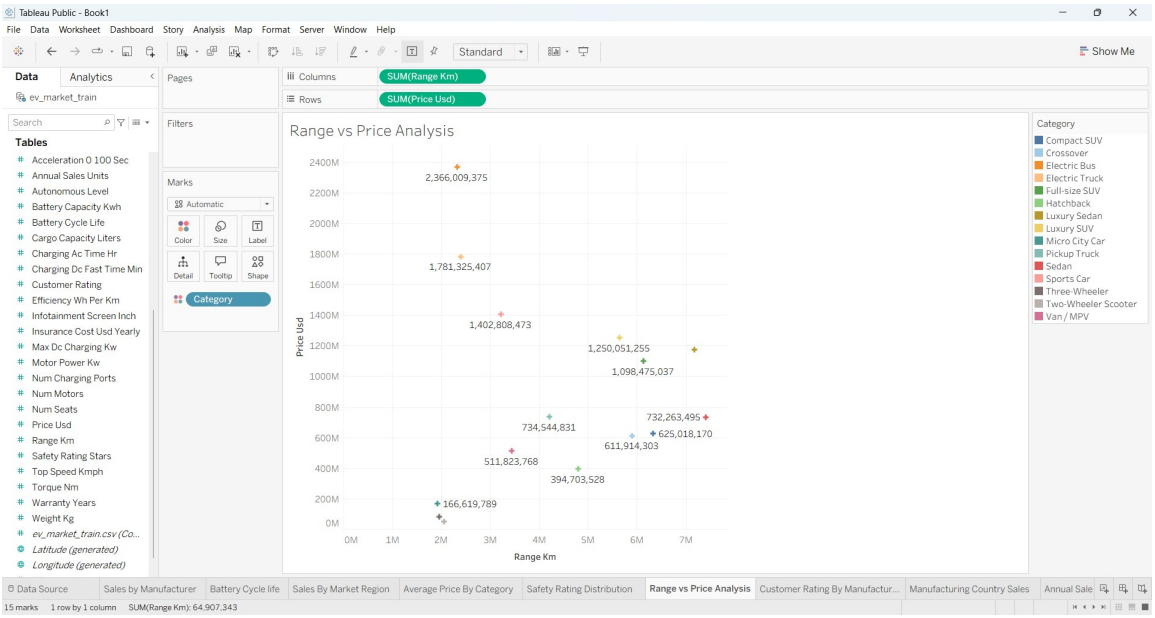


Safety Rating Distribution (Tableau worksheet)

6. Does paying more actually buy more range?

Visualization: Scatter plot of Range (X) against Price (Y), one point per category, coloured by Category.

Finding: Yes, but not in a straight line. The linear correlation is only 0.28, but with log(price) it rises to 0.67, so each extra dollar buys less range at the top end.

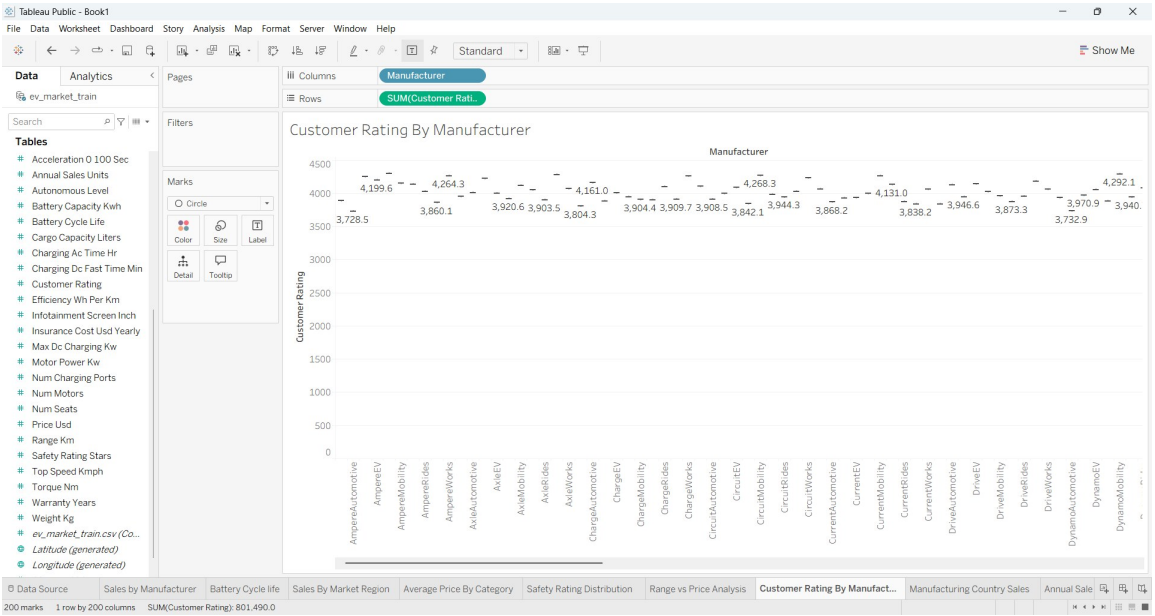


Range vs Price Analysis (Tableau worksheet)

7. Do customers rate some manufacturers much higher than others?

Visualization: Circle plot of Customer Rating by Manufacturer.

Finding: Average ratings only range from 4.02 (VoltAutomotive) to 4.14 (KineticAuto). Satisfaction is very uniform, so rating alone should not decide a purchase.

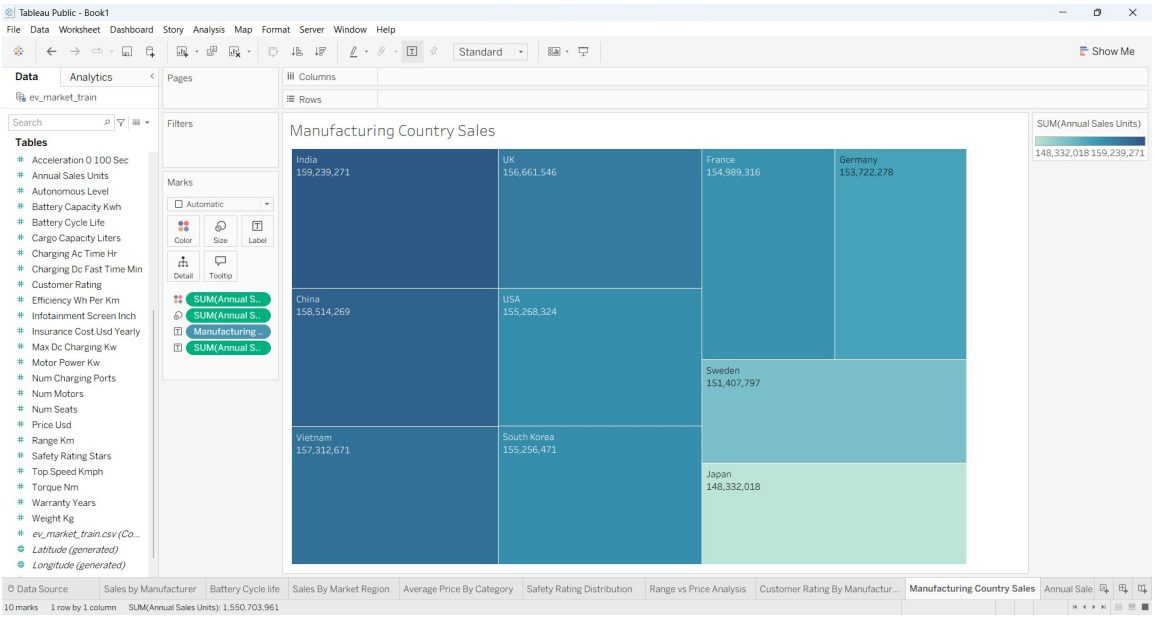


Customer Rating by Manufacturer (Tableau worksheet)

8. Which manufacturing countries contribute the most volume?

Visualization: Treemap sized and coloured by SUM(Annual Sales Units) per Manufacturing Country.

Finding: India (159.2M), China (158.5M) and Vietnam (157.3M) are the top three. Japan is lowest at 148.3M, which is only about 7% below India.

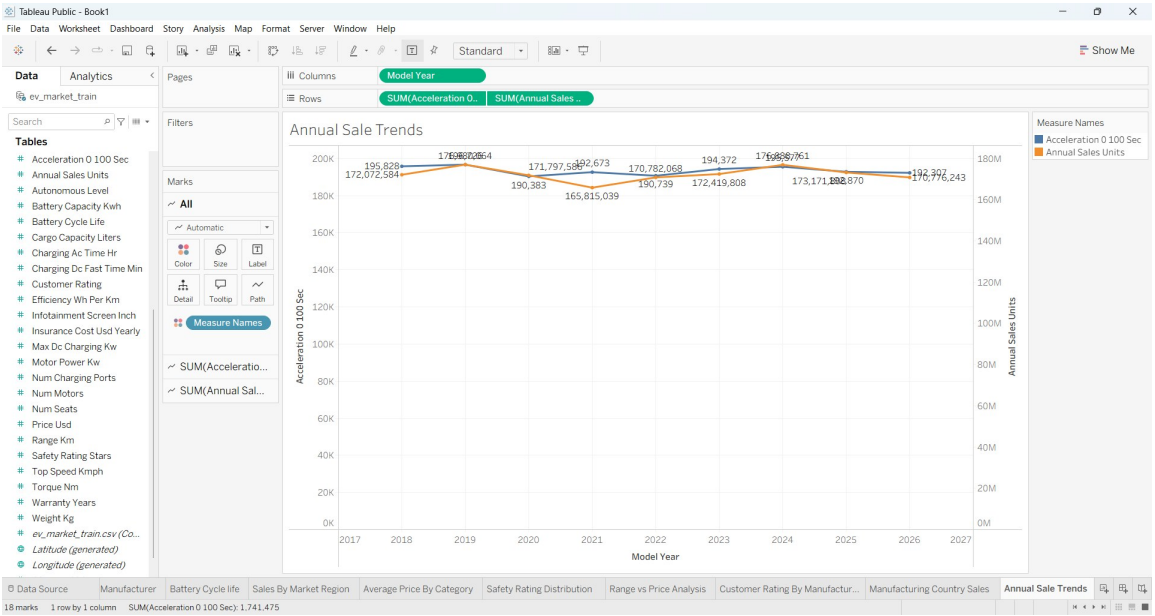


Manufacturing Country Sales (Tableau worksheet)

9. How have sales moved from model year 2018 to 2026?

Visualization: Line chart of annual sales by Model Year.

Finding: Sales peaked at 177.0M units in 2019 and 176.9M in 2024, with a low of 165.8M in 2021 (about 6% below the 2019 peak).



Annual Sale Trends (Tableau worksheet)